Nombre: Nathaly Bravo

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Métodos Númericos

Tarea 09

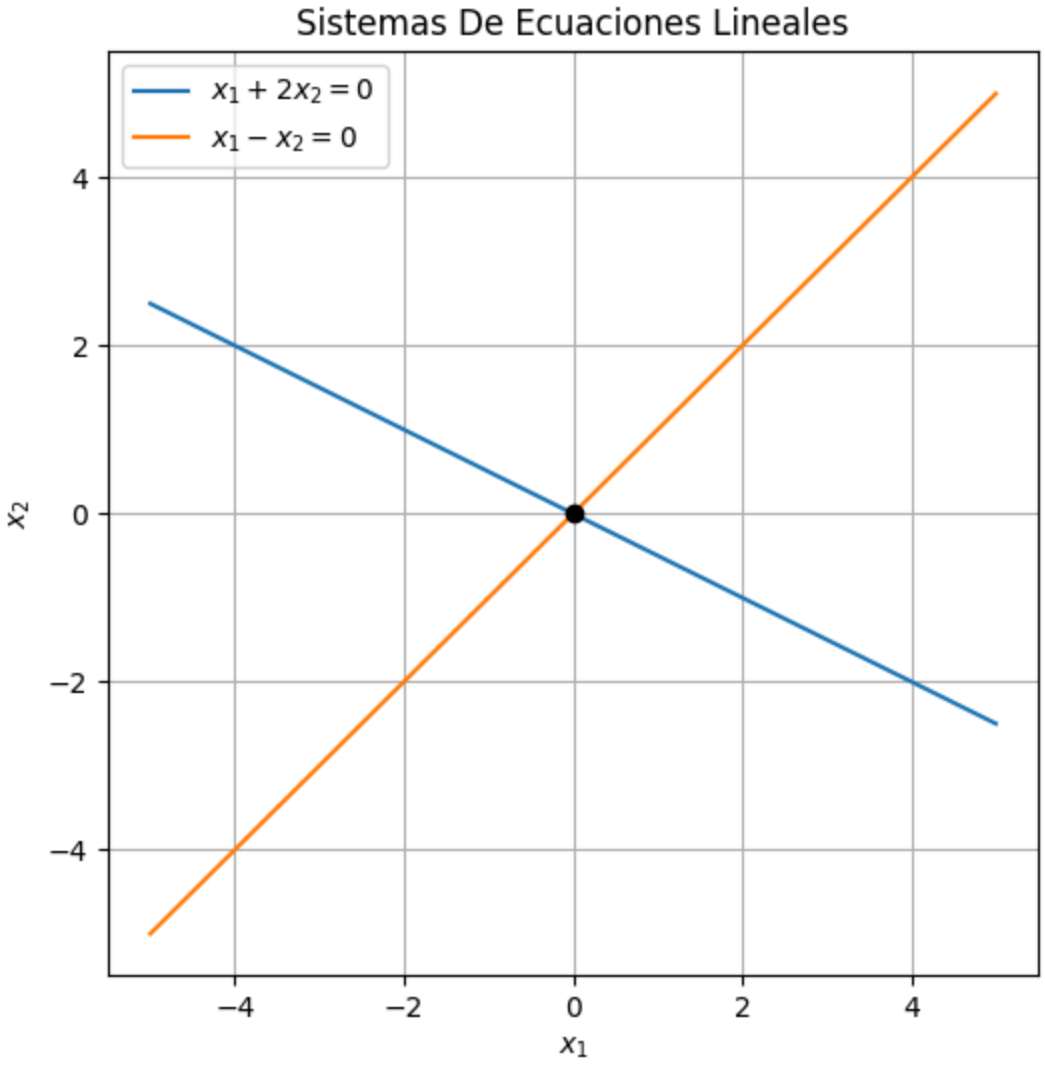
1)Para cada uno de los siguientes sistemas lineales, obtenga, de ser posible, una solución con métodos gráficos. Explique los resultados desde un punto de vista geométrico.

a)

$x_1 + 2x_2 = 0$

$x_1 - x_2$

In [3]:
```
import numpy as np
import matplotlib.pyplot as plt


x1 = np.linspace(-5, 5, 400)

x2_eq1 = -0.5 * x1
x2_eq2 = x1

plt.figure(figsize=(6,6))
plt.plot(x1, x2_eq1, label=r"$x_1 + 2x_2 = 0$")
plt.plot(x1, x2_eq2, label=r"$x_1 - x_2 = 0$")



plt.scatter(0, 0, color = "black", zorder=5)


plt.xlabel(r"$x_1$")
plt.ylabel(r"$x_2$")
plt.title("Sistemas De Ecuaciones Lineales")
plt.legend()
plt.grid(True)
plt.show()
```



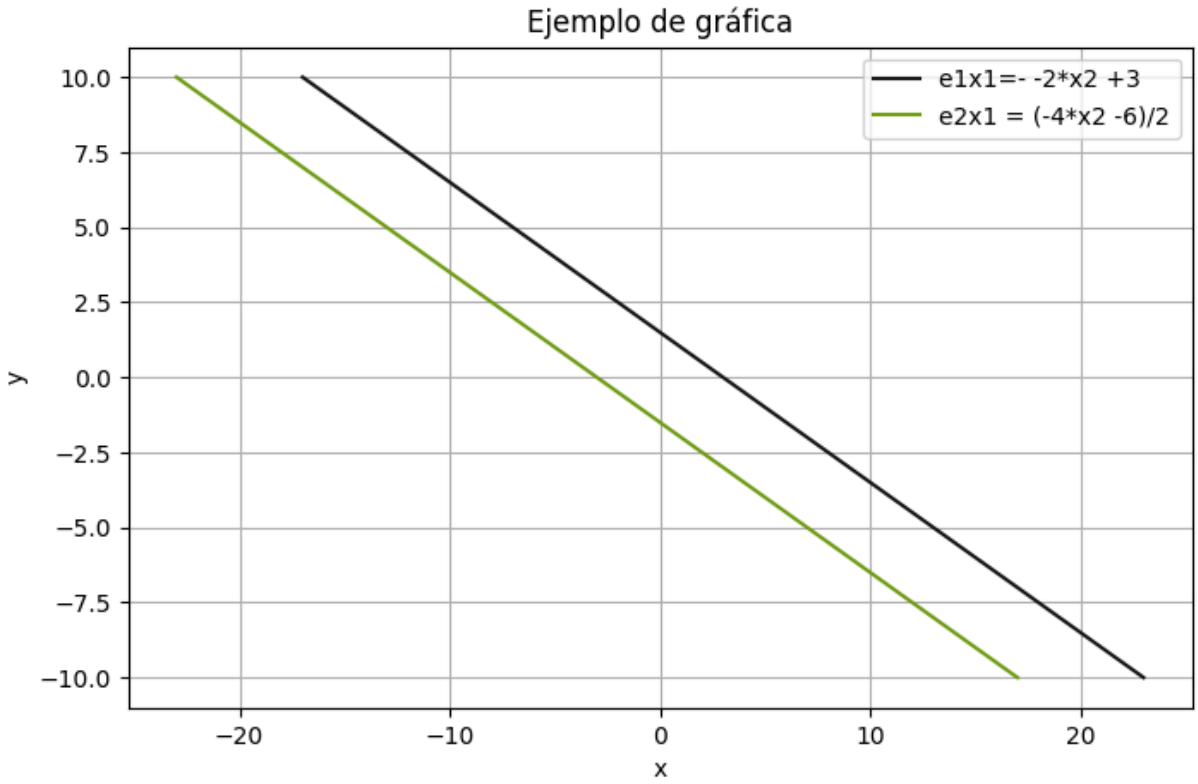
Respuesta : El sistema tiene una única solución, ya que las dos rectas se intersectan en un solo punto, el cual representa la solución del sistema de ecuaciones.

b.

$x_1 + 2x_2 = 3$

$-2x_1 - 4x_2 = 6$

In [4]:
```
x2 = np.linspace(-10, 10, 100)
x1e1= 3 - 2* x2
x1e2=  (-4*x2 -6)/2

plt.figure(figsize=(8, 5))
plt.plot(x1e1, x2,color="#141414", label='e1x1=- -2*x2 +3')
plt.plot(x1e2, x2, color="#6E9C12", label='e2x1 = (-4*x2 -6)/2')
plt.title('Ejemplo de gráfica')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



Respuestas: Ambas rectas no se intersectan en ningún punto, ya que son paralelas. Por lo tanto, el sistema no tiene solución.

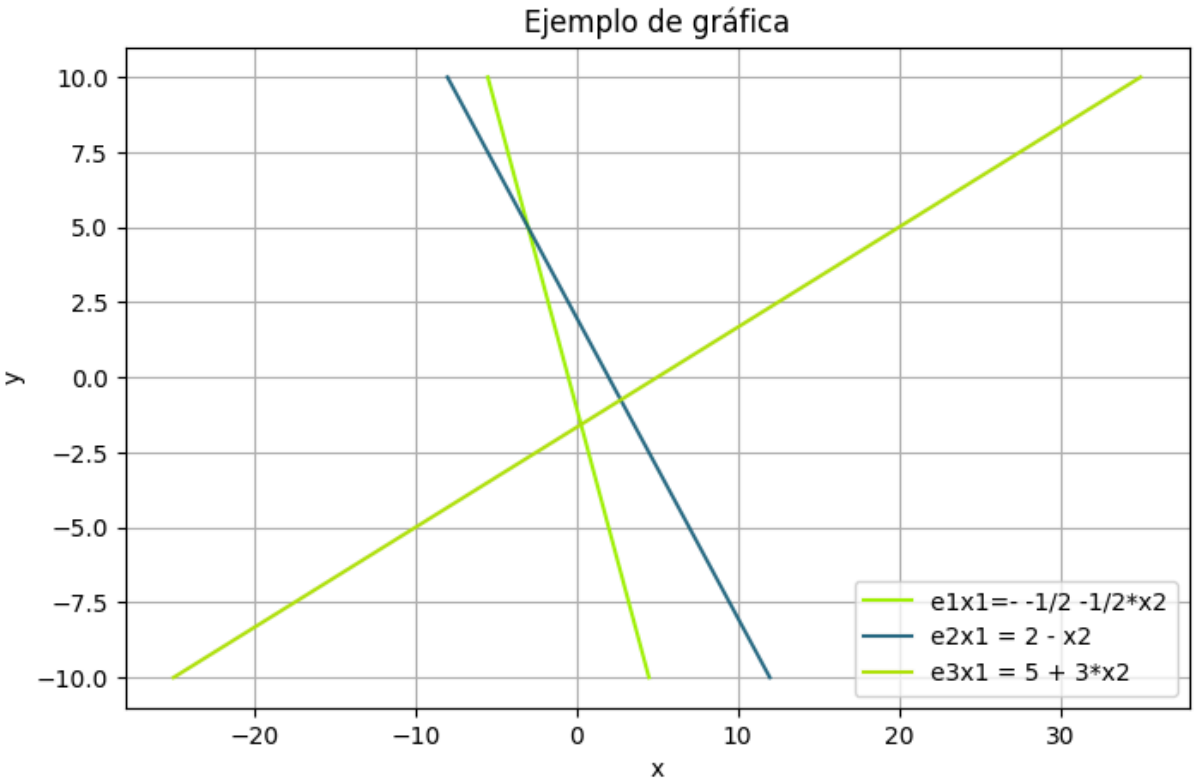
c.

$2x_1 + x_2 = -1$

$x_1 + x_2 = 2$

$x_1 - 3x_2 = 5$

In [5]:
```
x2 = np.linspace(-10, 10, 100)
x1e1= -1/2 -1/2*x2
x1e2=  2 - x2
x1e3= 5 + 3*x2

plt.figure(figsize=(8, 5))
plt.plot(x1e1, x2,color="#A0F405", label='e1x1=- -1/2 -1/2*x2')
plt.plot(x1e2, x2, color="#135D78E6", label='e2x1 = 2 - x2')
plt.plot(x1e3, x2, color="#B0E20C", label='e3x1 = 5 + 3*x2')
plt.title('Ejemplo de gráfica')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



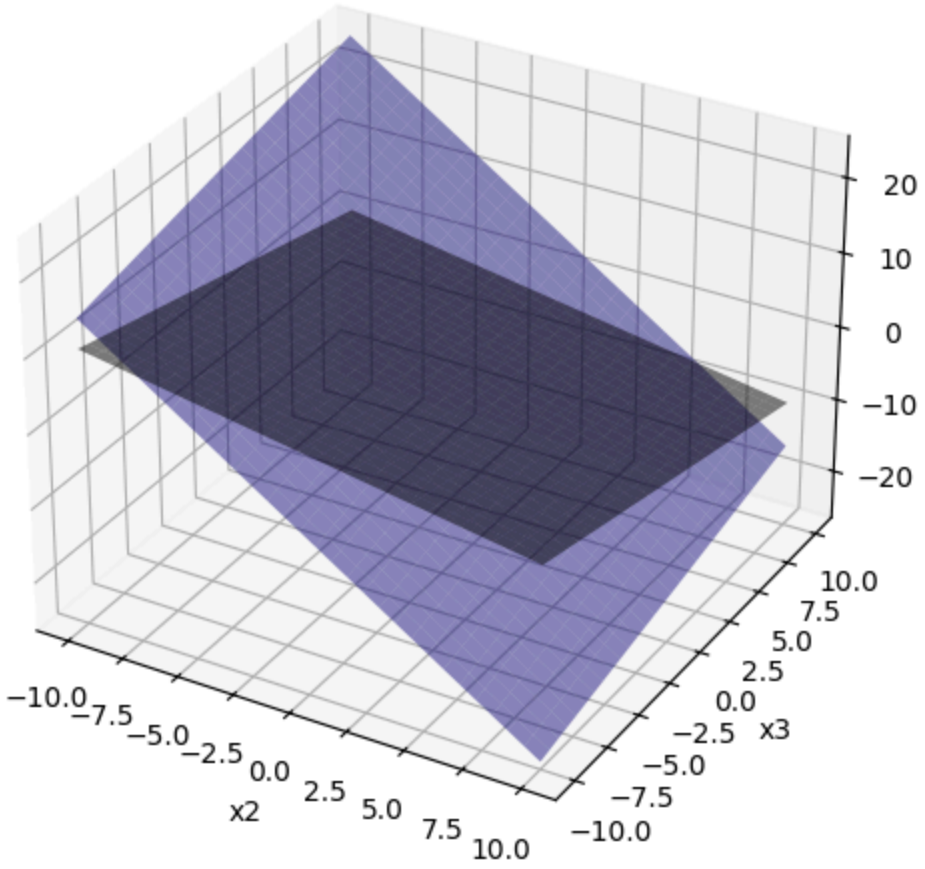
No existe un punto en el que las tres funciones itersecan por lo que el sistema no tiene solución,

d.

$2x_1 + x_2 + x_3 = 1$

$2x_1 + 4x_2 - x_3 = -1$

In [7]:
```
from mpl_toolkits.mplot3d import Axes3D

x2, x3 = np.meshgrid(np.linspace(-10,10,30), np.linspace(-10,10,30))

x1e1 = (1 - x2 - x3 ) * 0.5
x1e2 = (-1 - 4*x2 + x3) * 0.5

fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

ax.plot_surface(x2, x3, x1e1, alpha=0.5, color="#000000")
ax.plot_surface(x2, x3, x1e2, alpha=0.5, color="#3124E8"    )

ax.set_xlabel('x2')
ax.set_ylabel('x3')
ax.set_zlabel('x1')
plt.title('Sistema en 3D')
plt.show()
```



2) Utilice la eliminación gaussiana con sustitución hacia atrás y aritmética de redondeo de dos dígitos para resolver los siguientes sistemas lineales. No reordene las ecuaciones. (La solución exacta para cada sistema es $x1 = -1$

In [17]:
```python
import numpy as np
import logging
from sys import stdout
from datetime import datetime

logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def r(x):
    return np.round(x, 2)

def eliminacion_gaussiana(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """Resuelve un sistema de ecuaciones lineales mediante el método de eliminación

    ## Parameters
```

```
    ``A``: matriz aumentada del sistema de ecuaciones lineales. Debe ser de tamaño

    ## Return

    ``solucion``: vector con la solución del sistema de ecuaciones lineales.

    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)
    n = A.shape[0]

    for i in range(0, n - 1 ): #loop por columna

        # ---encontrar pivote
        p = None #default, first element
        for pi in range(i, n):
            if A[pi, i] == 0:
                # must be nonzero
                continue

            if p is None:
                # first nonzero element
                p = pi
                continue

            if abs(A[pi, i] < abs(A[p,i])):
                p = pi
        if p is None:
            # no pivot found.
            raise ValueError("No existe solución única.")

        if p != i:
            # swap rows
            logging.debug(f"Intercambiando filas {i} y {p}")
            _aux = A[i, :].copy()
            A[i, :] = A[p, :].copy()
            A[p, :] = _aux

        # --- Eliminación: loop por fila
        for j in range(i + 1, n):
            m = r( A[j, i] / A[i, i])
            A[j, i:] = r(A[j, i:] - r(m * A[i, i:]))

        logging.info(f"\n{A}")

    if A[n - 1, n - 1] == 0:
        raise ValueError("No existe solución única.")

        print(f"\n{A}")
    # --- Sustitución hacia atrás
    solucion = np.zeros(n)
    solucion[n - 1] =r( A[n - 1, n] / A[n - 1, n - 1])
```

```
    for i in range(n - 2, -1, -1):
        suma = 0
        for j in range(i + 1, n):
            suma += r(A[i, j] * solucion[j])
        solucion[i] = r((A[i, n] - suma) / A[i, i])

    return solucion
```

```
[07-08 18:48:59][INFO] 2026-07-08 18:48:59.810486
```

a.

$$-x_1 + 4x_2 + x_3 = 8$$

$$\frac{5}{3}x_1 + \frac{2}{3}x_2 + \frac{2}{3}x_3$$

$$2x_1 + x_2 + 4x_3 = 11$$

In [18]:
```
Aa = [
    [-1., 4, 1, 8],
    [5/3, 2/3, 2/3, 1],
    [2, 1, 4, 11]
]

eliminacion_gaussiana(Aa)
```

```
[07-08 18:49:03][INFO]
[[-1.    4.    1.    8.  ]
 [-0.    7.35  2.34 14.36]
 [ 0.    9.    6.   27.  ]]
[07-08 18:49:03][INFO]
[[-1.    4.    1.    8.  ]
 [-0.    7.35  2.34 14.36]
 [ 0.    0.03  3.15  9.48]]
```

Out[18]:
```
array([-0.99,  1.  ,  3.01])
```

b.

$$4x_1 + 2x_2 - x_3 = -5$$

$$\frac{1}{9}x_1 + \frac{1}{9}x_2 - \frac{1}{3}x_3 = -1$$

$$x_1 + 4x_2 + 2x_3$$

In [19]:
```
Ab = [
    [4., 2,-1,-5],
    [1/9,1/9,-1/3,-1],
    [1,4,2,9]
]
eliminacion_gaussiana(Ab)
```

```
[07-08 18:49:06][INFO]
[[ 0.11111111  0.11111111 -0.33333333 -1.        ]
 [ 0.         -2.         11.         31.        ]
 [ 0.          3.          5.         18.        ]]
[07-08 18:49:06][INFO]
[[ 0.11111111  0.11111111 -0.33333333 -1.        ]
 [ 0.         -2.         11.         31.        ]
 [ 0.          0.         21.5        64.5       ]]
```

Out[19]: array([-0.99, 1. , 3.])

3) Utilice el algoritmo de eliminación gaussiana para resolver, de ser posible, los siguientes sistemas lineales, y determine si se necesitan intercambios de fila

In [21]:

```python
import numpy as np
import logging
from sys import stdout
from datetime import datetime

intercambio=0
logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def eliminacion_gaussiana(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """Resuelve un sistema de ecuaciones lineales mediante el método de eliminación

    ## Parameters

    ``A``: matriz aumentada del sistema de ecuaciones lineales. Debe ser de tamaño

    ## Return

    ``solucion``: vector con la solución del sistema de ecuaciones lineales.

    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)
    n = A.shape[0]

    for i in range(0, n - 1):  # loop por columna

        # --- encontrar pivote
        p = None  # default, first element
        for pi in range(i, n):
            if A[pi, i] == 0:
                # must be nonzero
                continue
```

```
            if p is None:
                # first nonzero element
                p = pi
                continue

            if abs(A[pi, i]) < abs(A[p, i]):
                p = pi

        if p is None:
            # no pivot found.
            raise ValueError("No existe solución única.")

        if p != i:
            # swap rows
            logging.debug(f"Intercambiando filas {i} y {p}")
            _aux = A[i, :].copy()
            A[i, :] = A[p, :].copy()
            A[p, :] = _aux
            global intercambio
            intercambio+=1

        # --- Eliminación: loop por fila
        for j in range(i + 1, n):
            m = A[j, i] / A[i, i]
            A[j, i:] = A[j, i:] - m * A[i, i:]

        logging.info(f"\n{A}")

    if A[n - 1, n - 1] == 0:
        print( ValueError("No existe solución única."))
        return

        print(f"\n{A}")
    # --- Sustitución hacia atrás
    solucion = np.zeros(n)
    solucion[n - 1] = A[n - 1, n] / A[n - 1, n - 1]

    for i in range(n - 2, -1, -1):
        suma = 0
        for j in range(i + 1, n):
            suma += A[i, j] * solucion[j]
        solucion[i] = (A[i, n] - suma) / A[i, i]

    print("Número de intercambios de filas:", intercambio)
    return solucion
```

```
[07-08 18:49:29][INFO] 2026-07-08 18:49:29.228684
```

a.

$$x_1 - x_2 + 3x_3 = 2$$
$$3x_1 - 3x_2 + x_3 = -1$$
$$x_1 + x_2 = 3$$

In [22]:
```
A_a = [
    [1.,-1,3,2],
    [3,-3,1,-1],
    [1,1,0,3]
]

eliminacion_gaussiana(A_a)
```

```
[07-08 18:49:32][INFO]
[[ 1. -1.  3.  2.]
 [ 0.  0. -8. -7.]
 [ 0.  2. -3.  1.]]
[07-08 18:49:32][INFO]
[[ 1. -1.  3.  2.]
 [ 0.  2. -3.  1.]
 [ 0.  0. -8. -7.]]
Número de intercambios de filas: 1
```

Out[22]:
```
array([1.1875, 1.8125, 0.875 ])
```

b.

$$2x_1 - 1.5x_2 + 3x_3 = 1$$
$$-x_1 + 2x_3 = 3$$
$$4x_1 - 4.5x_2 + 5x_3 = 1$$

In [23]:
```
A_b = [[2., -1/5, 3,1], [-1,0 , 2, 3], [4,-4.5, 5, 1]]

eliminacion_gaussiana(A_b)
```

```
[07-08 18:49:34][INFO]
[[-1.   0.   2.   3. ]
 [ 0.  -0.2  7.   7. ]
 [ 0.  -4.5 13.  13. ]]
[07-08 18:49:34][INFO]
[[  -1.     0.     2.     3. ]
 [   0.    -0.2    7.     7. ]
 [   0.     0.  -144.5 -144.5]]
Número de intercambios de filas: 2
```

Out[23]:
```
array([-1., -0.,  1.])
```

c.

$$2x_1 = 3$$
$$x_1 + 1.5x_2 = 4.5$$
$$-3x_2 + 0.5x_3 = -6.6$$
$$2x_1 - 2x_2 + x_3 + x_4 = 0.8$$

In [24]:
```
A_c = [
    [2.,0,0,0,3],
    [1,1.5,0,0,4.5],
    [0,-3,0.5,0,-6.6],
    [2,-2,1,1,0.8]
]
eliminacion_gaussiana(A_c)
```

```
[07-08 18:49:37][INFO]
[[ 1.   1.5  0.   0.   4.5]
 [ 0.  -3.   0.   0.  -6. ]
 [ 0.  -3.   0.5  0.  -6.6]
 [ 0.  -5.   1.   1.  -8.2]]
[07-08 18:49:37][INFO]
[[ 1.   1.5  0.   0.   4.5]
 [ 0.  -3.   0.   0.  -6. ]
 [ 0.   0.   0.5  0.  -0.6]
 [ 0.   0.   1.   1.   1.8]]
[07-08 18:49:37][INFO]
[[ 1.   1.5  0.   0.   4.5]
 [ 0.  -3.   0.   0.  -6. ]
 [ 0.   0.   0.5  0.  -0.6]
 [ 0.   0.   0.   1.   3. ]]
Número de intercambios de filas: 3
```

Out[24]: array([1.5, 2. , -1.2, 3.])

d.

x_1 + x_2 + x_4 = 2

$$2x_1 + x_2 - x_3 + x_4 = 1$$
$$4x_1 - x_2 - 2x_3 + 2x_4 = 0$$
$$3x_1 - x_2 - x_3 + 2x_4 = -3$$

In [25]:
```
A_d = [
    [1,1,0,1,2],
    [2,1,-1,1,1],
    [4,-1,-2,2,0],
    [3,-1,-1,2,-3]
]

eliminacion_gaussiana(A_d)
```

```
[07-08 18:49:40][INFO]
[[ 1  1  0  1  2]
 [ 0 -1 -1 -1 -3]
 [ 0 -5 -2 -2 -8]
 [ 0 -4 -1 -1 -9]]
[07-08 18:49:40][INFO]
[[ 1  1  0  1  2]
 [ 0 -1 -1 -1 -3]
 [ 0  0  3  3  7]
 [ 0  0  3  3  3]]
[07-08 18:49:40][INFO]
[[ 1  1  0  1  2]
 [ 0 -1 -1 -1 -3]
 [ 0  0  3  3  7]
 [ 0  0  0  0 -4]]
No existe solución única.
```

4) Use el algoritmo de eliminación gaussiana y la aritmética computacional de precisión de 32 bits para resolver los siguientes sistemas lineales.

In [26]:

```python
import numpy as np
import logging
from sys import stdout
from datetime import datetime

intercambio=0
logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def eliminacion_gaussiana(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """Resuelve un sistema de ecuaciones lineales mediante el método de eliminación

    ## Parameters

    ``A``: matriz aumentada del sistema de ecuaciones lineales. Debe ser de tamaño

    ## Return

    ``solucion``: vector con la solución del sistema de ecuaciones lineales.

    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A , dtype=np.float32)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)
    n = A.shape[0]

    for i in range(0, n - 1):  # loop por columna

        # --- encontrar pivote
        p = None  # default, first element
        for pi in range(i, n):
            if A[pi, i] == 0:
                # must be nonzero
                continue

            if p is None:
                # first nonzero element
                p = pi
                continue

            if abs(A[pi, i]) < abs(A[p, i]):
                p = pi

        if p is None:
            # no pivot found.
            raise ValueError("No existe solución única.")

        if p != i:
            # swap rows
```

```
            logging.debug(f"Intercambiando filas {i} y {p}")
            _aux = A[i, :].copy()
            A[i, :] = A[p, :].copy()
            A[p, :] = _aux
            global intercambio
            intercambio+=1

        # --- Eliminación: loop por fila
        for j in range(i + 1, n):
            m = A[j, i] / A[i, i]
            A[j, i:] = A[j, i:] - m * A[i, i:]

        logging.info(f"\n{A}")

    if A[n - 1, n - 1] == 0:
        raise ValueError("No existe solución única.")

        print(f"\n{A}")
    # --- Sustitución hacia atrás
    solucion = np.zeros(n)
    solucion[n - 1] = A[n - 1, n] / A[n - 1, n - 1]

    for i in range(n - 2, -1, -1):
        suma = 0
        for j in range(i + 1, n):
            suma += A[i, j] * solucion[j]
        solucion[i] = (A[i, n] - suma) / A[i, i]

    print("Número de intercambios de filas:", intercambio)
    return solucion


Ab = [[1/4, 1/5, 1/6, 9], [1/3, 1/4, 1/5, 8], [1/2 ,1, 2, 8]]

eliminacion_gaussiana(Ab)
```

```
[07-08 18:49:52][INFO] 2026-07-08 18:49:52.315885
[07-08 18:49:52][INFO]
[[  0.25         0.2          0.16666667   9.        ]
 [  0.          -0.01666668  -0.02222224  -4.        ]
 [  0.           0.6          1.6666666  -10.        ]]
[07-08 18:49:52][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  9.0000000e+00]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -4.0000000e+00]
 [ 0.0000000e+00 -5.9604645e-08  8.6666673e-01 -1.5399989e+02]]
Número de intercambios de filas: 0
```

Out[26]: array([-227.07666725, 476.92263902, -177.69216919])

a.

$$\frac{1}{4}x_1 + \frac{1}{5}x_2 + \frac{1}{6}x_3 = 9$$

$$\frac{1}{3}x_1 + \frac{1}{4}x_2 + \frac{1}{5}x_3 = 8$$

$$\frac{1}{2}x_1 + x_2 + 2x_3 = 8$$

In [27]:
```
A_aa = [
    [1/4,1/5,1/6,9],
    [1/3,1/4,1/5,8],
    [1/2,1,2,8]
]

eliminacion_gaussiana(A_aa)
```

```
[07-08 18:49:56][INFO]
[[  0.25         0.2          0.16666667   9.        ]
 [  0.          -0.01666668  -0.02222224  -4.        ]
 [  0.           0.6          1.6666666  -10.        ]]
[07-08 18:49:56][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  9.0000000e+00]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -4.0000000e+00]
 [ 0.0000000e+00 -5.9604645e-08  8.6666673e-01 -1.5399989e+02]]
Número de intercambios de filas: 0
```

Out[27]:
```
array([-227.07666725,  476.92263902, -177.69216919])
```

b.

$$3.333x_1 + 15920x_2 - 10.333x_3 = 15913$$

$$2.222x_1 + 16.71x_2 + 9.612x_3 = 28.544$$

$$1.5611x_1 + 5.1791x_2 + 1.6852x_3 = 8,4254$$

In [28]:
```
A_ab = [
    [3.333,15920,-10.333,15913],
    [2.222,16.71,9.612,28.544],
    [1.5611,5.1791,1.6852,8.4254]
]

eliminacion_gaussiana(A_ab)
```

```
[07-08 18:49:59][INFO]
[[ 1.5611000e+00  5.1791000e+00  1.6852000e+00  8.4253998e+00]
 [ 0.0000000e+00  9.3383007e+00  7.2133622e+00  1.6551662e+01]
 [ 0.0000000e+00  1.5908942e+04 -1.3930958e+01  1.5895012e+04]]
[07-08 18:49:59][INFO]
[[ 1.5611000e+00  5.1791000e+00  1.6852000e+00  8.4253998e+00]
 [ 0.0000000e+00  9.3383007e+00  7.2133622e+00  1.6551662e+01]
 [ 0.0000000e+00  0.0000000e+00 -1.2302779e+04 -1.2302777e+04]]
Número de intercambios de filas: 1
```

Out[28]:
```
array([0.99999975, 1.00000009, 0.99999982])
```

c.

$$x_1 + \frac{1}{2}x_2 + \frac{1}{3}x_3 + \frac{1}{4}x_4 = \frac{1}{6}$$

$$\frac{1}{2}x_1 + \frac{1}{3}x_2 + \frac{1}{4}x_3 + \frac{1}{5}x_4 = \frac{1}{7}$$

$$\frac{1}{3}x_1 + \frac{1}{4}x_2 + \frac{1}{5}x_3 + \frac{1}{6}x_4 = \frac{1}{8}$$

$$\frac{1}{4}x_1 + \frac{1}{5}x_2 + \frac{1}{6}x_3 + \frac{1}{7}x_4 = \frac{1}{9}$$

In [29]:
```
A_ac = [
    [1,1/2,1/3,1/4,1/6],
    [1/2,1/3,1/4,1/5,1/7],
    [1/3,1/4,1/5,1/6,1/8],
    [1/4,1/5,1/6,1/7,1/9]
]

eliminacion_gaussiana(A_ac)
```

```
[07-08 18:50:02][INFO]
[[ 0.25        0.2         0.16666667  0.14285715  0.11111111]
 [ 0.         -0.06666666 -0.08333334 -0.0857143  -0.07936507]
 [ 0.         -0.01666668 -0.02222224 -0.02380954 -0.02314815]
 [ 0.         -0.3        -0.33333334 -0.3214286  -0.2777778 ]]
[07-08 18:50:02][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  1.4285715e-01
   1.1111111e-01]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -2.3809537e-02
  -2.3148149e-02]
 [ 0.0000000e+00  0.0000000e+00  5.5555180e-03  9.5237717e-03
   1.3227440e-02]
 [ 0.0000000e+00  2.9802322e-08  6.6666603e-02  1.0714275e-01
   1.3888860e-01]]
[07-08 18:50:02][INFO]
[[ 2.5000000e-01  2.0000000e-01  1.6666667e-01  1.4285715e-01
   1.1111111e-01]
 [ 0.0000000e+00 -1.6666681e-02 -2.2222236e-02 -2.3809537e-02
  -2.3148149e-02]
 [ 0.0000000e+00  0.0000000e+00  5.5555180e-03  9.5237717e-03
   1.3227440e-02]
 [ 0.0000000e+00  2.9802322e-08  0.0000000e+00 -7.1431771e-03
  -1.9841611e-02]]
Número de intercambios de filas: 3
```

Out[29]: array([-0.03174083, 0.5951854 , -2.3808313 , 2.77770114])

d.

$$2x_1 + x_2 - x_3 + x_4 - 3x_5 = 7$$

$$x_1 + 2x_3 - x_4 + x_5 = 2$$

$$-2x_2 - x_3 + x_4 - x_5 = -5$$

$$3x_1 + x_2 - 4x_3 + 5x_5 = 6$$

$$x_1 - x_2 - x_3 - x_4 + x_5 = -3$$

In [30]:
```
A_ad = [
    [2.,1,-1,1,-3,7],
    [1,0,2,-1,1,2],
    [0,-2,-1,1,-1,-5],
    [3,1,-4,0,5,6],
    [1,-1,-1,-1,1,-3]
]

eliminacion_gaussiana(A_ad)
```

```
[07-08 18:50:05][INFO]
[[  1.   0.   2.  -1.   1.   2.]
 [  0.   1.  -5.   3.  -5.   3.]
 [  0.  -2.  -1.   1.  -1.  -5.]
 [  0.   1. -10.   3.   2.   0.]
 [  0.  -1.  -3.   0.   0.  -5.]]
[07-08 18:50:05][INFO]
[[  1.   0.   2.  -1.   1.   2.]
 [  0.   1.  -5.   3.  -5.   3.]
 [  0.   0. -11.   7. -11.   1.]
 [  0.   0.  -5.   0.   7.  -3.]
 [  0.   0.  -8.   3.  -5.  -2.]]
[07-08 18:50:05][INFO]
[[  1.          0.          2.         -1.          1.          2.       ]
 [  0.          1.         -5.          3.         -5.          3.       ]
 [  0.          0.         -5.          0.          7.         -3.       ]
 [  0.          0.          0.          7.        -26.400002    7.6000004]
 [  0.          0.          0.          3.        -16.2         2.8000002]]
[07-08 18:50:05][INFO]
[[  1.          0.          2.         -1.          1.          2.       ]
 [  0.          1.         -5.          3.         -5.          3.       ]
 [  0.          0.         -5.          0.          7.         -3.       ]
 [  0.          0.          0.          3.        -16.2         2.8000002]
 [  0.          0.          0.          0.         11.399998    1.0666666]]
Número de intercambios de filas: 6
```

Out[30]:
```
array([1.88304105, 2.80701722, 0.73099417, 1.43859666, 0.09356727])
```

5)Dado el sistema lineal:

$$x_1 - x_2 + \alpha x_3 = -2$$

$$-x_1 + 2x_2 - \alpha x_3 = 3$$

$$\alpha x_1 + x_2 + x3 = 2$$

a. Encuentre el valor(es) de α para los que el sistema no tiene soluciones.

b. Encuentre el valor(es) de α para los que el sistema tiene un número infinito de soluciones.

c. Suponga que existe una única solución para una a determinada, encuentre la solución.

Ejercicios aplicados

In [34]:
```python
import sympy as sp


alpha = sp.Symbol('alpha')


A = sp.Matrix([
    [1, -1,  alpha],
    [-1, 2, -alpha],
    [alpha, 1, 1]
])

b = sp.Matrix([-2, 3, 2])

# Matriz ampliada [A | b]
Ab = A.row_join(b)

# --- Determinante en funcion de alpha ---
detA = sp.simplify(A.det())
print("det(A) =", detA)

# Valores de alpha que anulan el determinante (casos criticos)
alphas_criticos = sp.solve(detA, alpha)
print("alpha que anulan det(A):", alphas_criticos)
```

```
det(A) = 1 - alpha**2
alpha que anulan det(A): [-1, 1]
```

In [ ]:
```python
for val in alphas_criticos:
    A_val  = A.subs(alpha, val)
    Ab_val = Ab.subs(alpha, val)

    rango_A  = A_val.rank()
    rango_Ab = Ab_val.rank()

    print(f"\nalpha = {val}")
    print(f"  rango(A)      = {rango_A}")
    print(f"  rango([A|b])  = {rango_Ab}")

    if rango_A < rango_Ab:
        print("  --> SIN solucion (sistema incompatible)")
    elif rango_A == rango_Ab < A.shape[1]:
        print("  --> INFINITAS soluciones")
    else:
        print("  --> solucion unica")
```

```
alpha = -1
  rango(A)      = 2
  rango([A|b])  = 2
  --> INFINITAS soluciones

alpha = 1
  rango(A)      = 2
  rango([A|b])  = 3
  --> SIN solucion (sistema incompatible)
```

In [ ]:
```python
val = 0

A_val = A.subs(alpha, val)
b_val = b

x = A_val.solve(b_val)   # resuelve A x = b
print(f"\nSolucion para alpha = {val}:")
print("x1, x2, x3 =", x.T)
```

```
Solucion para alpha = 0:
x1, x2, x3 = Matrix([[-1, 1, 1]])
```

7) Repita el ejercicio 4 con el método Gauss-Jordan.

In [71]:
```python
import numpy as np
import logging
from sys import stdout
from datetime import datetime

logging.basicConfig(
    level=logging.INFO,
    format="[%(asctime)s][%(levelname)s] %(message)s",
    stream=stdout,
    datefmt="%m-%d %H:%M:%S",
)
logging.info(datetime.now())

def gauss_jordan(A: np.ndarray | list[list[float | int]]) -> np.ndarray:
    """
    Resuelve un sistema de ecuaciones lineales mediante el método de Gauss-Jordan.

    Parameters
    ----------
    A : matriz aumentada del sistema de ecuaciones lineales (n x n+1)

    Returns
    -------
    solucion : vector con la solución del sistema de ecuaciones lineales
    """
    if not isinstance(A, np.ndarray):
        logging.debug("Convirtiendo A a numpy array.")
        A = np.array(A, dtype=np.float32)
    assert A.shape[0] == A.shape[1] - 1, "La matriz A debe ser de tamaño n-by-(n+1)"
    n = A.shape[0]
```

```
    for i in range(n):
        # Pivoteo parcial si es necesario
        max_row = np.argmax(np.abs(A[i:, i])) + i
        if A[max_row, i] == 0:
            raise ValueError("No existe solución única.")
        if max_row != i:
            A[[i, max_row]] = A[[max_row, i]]

        # Normalizar la fila pivote
        A[i] = A[i] / A[i, i]

        # Hacer ceros en todas las filas excepto la pivote
        for j in range(n):
            if j != i:
                A[j] = A[j] - A[j, i] * A[i]

        logging.info(f"\n{A}")

    solucion = A[:, -1]
    return solucion
```

```
[01-07 21:51:38][INFO] 2026-01-07 21:51:38.991521
```

a.

In [72]:
```
A_aaa = [
    [1/4,1/5,1/6,9],
    [1/3,1/4,1/5,8],
    [1/2,1,2,8]
]

gauss_jordan(A_aaa)
```

```
[01-07 21:51:39][INFO]
[[ 1.         2.         4.        16.       ]
 [ 0.        -0.4166667 -1.1333333  2.6666665]
 [ 0.        -0.3       -0.8333333  5.       ]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00 -1.4399996e+00  2.8799999e+01]
 [-0.0000000e+00  1.0000000e+00  2.7199998e+00 -6.3999991e+00]
 [ 0.0000000e+00  0.0000000e+00 -1.7333329e-02  3.0800002e+00]]
[01-07 21:51:39][INFO]
[[   1.         0.         0.      -227.07693]
 [   0.         1.         0.       476.92322]
 [  -0.        -0.         1.      -177.69237]]
```

Out[72]:
```
array([-227.07693,  476.92322, -177.69237], dtype=float32)
```

b.

In [73]:
```
A_abb = [
    [3.333,15920,-10.333,15913],
    [2.222,16.71,9.612,28.544],
    [1.5611,5.1791,1.6852,8.4254]
]
```

```
gauss_jordan(A_abb)
```

```
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  4.7764775e+03 -3.1002102e+00  4.7743774e+03]
 [ 0.0000000e+00 -1.0596623e+04  1.6500668e+01 -1.0580122e+04]
 [ 0.0000000e+00 -7.4513799e+03  6.5249386e+00 -7.4448555e+03]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00  4.3375430e+00  5.3378906e+00]
 [-0.0000000e+00  1.0000000e+00 -1.5571628e-03  9.9844283e-01]
 [ 0.0000000e+00  0.0000000e+00 -5.0780735e+00 -5.0786133e+00]]
[01-07 21:51:39][INFO]
[[ 1.         0.         0.         0.9998865]
 [-0.         1.         0.         1.0000001]
 [-0.        -0.         1.         1.0001063]]
```

Out[73]: array([0.9998865, 1.0000001, 1.0001063], dtype=float32)

c.

In [74]:
```
A_acc = [
    [1,1/2,1/3,1/4,1/6],
    [1/2,1/3,1/4,1/5,1/7],
    [1/3,1/4,1/5,1/6,1/8],
    [1/4,1/5,1/6,1/7,1/9]
]

gauss_jordan(A_acc)
```

```
[01-07 21:51:39][INFO]
[[1.         0.5        0.33333334 0.25       0.16666667]
 [0.         0.08333334 0.08333333 0.075      0.05952381]
 [0.         0.08333333 0.08888888 0.08333334 0.06944444]
 [0.         0.075      0.08333334 0.08035715 0.06944445]]
[01-07 21:51:39][INFO]
[[ 1.          0.         -0.16666657 -0.19999996 -0.19047616]
 [ 0.          1.          0.9999998   0.8999999   0.7142857 ]
 [ 0.          0.          0.00555557  0.00833335  0.00992064]
 [ 0.          0.          0.00833335  0.01285715  0.01587302]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00  0.0000000e+00  5.7142526e-02
   1.2698348e-01]
 [ 0.0000000e+00  1.0000000e+00  0.0000000e+00 -6.4285570e-01
  -1.1904728e+00]
 [ 0.0000000e+00  0.0000000e+00  1.0000000e+00  1.5428559e+00
   1.9047589e+00]
 [ 0.0000000e+00  0.0000000e+00  0.0000000e+00 -2.3809634e-04
  -6.6138338e-04]]
[01-07 21:51:39][INFO]
[[ 1.          0.          0.          0.         -0.03174686]
 [ 0.          1.          0.          0.          0.5952499 ]
 [ 0.          0.          1.          0.         -2.380982  ]
 [-0.         -0.         -0.          1.          2.7777972 ]]
```

Out[74]: array([-0.03174686, 0.5952499 , -2.380982 , 2.7777972], dtype=float32)

d.

In [75]:
```
A_add = [
    [2.,1,-1,1,-3,7],
    [1,0,2,-1,1,2],
    [0,-2,-1,1,-1,-5],
    [3,1,-4,0,5,6],
    [1,-1,-1,-1,1,-3]
]

gauss_jordan(A_add)
```

```
[01-07 21:51:39][INFO]
[[ 1.          0.33333334 -1.3333334   0.          1.6666666   2.        ]
 [ 0.         -0.33333334  3.3333335  -1.         -0.6666666   0.        ]
 [ 0.         -2.         -1.          1.         -1.         -5.        ]
 [ 0.          0.3333333   1.6666667   1.         -6.333333    3.        ]
 [ 0.         -1.3333334   0.33333337 -1.         -0.6666666  -5.        ]]
[01-07 21:51:39][INFO]
[[ 1.0000000e+00  0.0000000e+00 -1.5000000e+00  1.6666667e-01
   1.5000000e+00  1.1666666e+00]
 [-0.0000000e+00  1.0000000e+00  5.0000000e-01 -5.0000000e-01
   5.0000000e-01  2.5000000e+00]
 [ 0.0000000e+00  0.0000000e+00  3.5000002e+00 -1.1666666e+00
  -4.9999994e-01  8.3333337e-01]
 [ 0.0000000e+00  0.0000000e+00  1.5000001e+00  1.1666666e+00
  -6.4999995e+00  2.1666667e+00]
 [ 0.0000000e+00  0.0000000e+00  1.0000000e+00 -1.6666667e+00
   5.9604645e-08 -1.6666665e+00]]
[01-07 21:51:39][INFO]
[[ 1.          0.          0.         -0.3333333   1.2857144   1.5238094 ]
 [-0.          1.          0.         -0.33333334  0.57142854  2.3809524 ]
 [ 0.          0.          1.         -0.3333333  -0.14285712  0.23809524]
 [ 0.          0.          0.          1.6666666  -6.2857137   1.8095238 ]
 [ 0.          0.          0.         -1.3333335   0.14285718 -1.9047618 ]]
[01-07 21:51:39][INFO]
[[ 1.          0.          0.          0.          0.02857172  1.8857142 ]
 [ 0.          1.          0.          0.         -0.68571424  2.7428572 ]
 [ 0.          0.          1.          0.         -1.3999997   0.6       ]
 [ 0.          0.          0.          1.         -3.7714283   1.0857143 ]
 [ 0.          0.          0.          0.         -4.8857145  -0.45714247]]
[01-07 21:51:39][INFO]
[[ 1.          0.          0.          0.          0.          1.8830408 ]
 [ 0.          1.          0.          0.          0.          2.8070176 ]
 [ 0.          0.          1.          0.          0.          0.73099405]
 [ 0.          0.          0.          1.          0.          1.4385962 ]
 [-0.         -0.         -0.         -0.          1.          0.09356717]]
```

Out[75]:
```
array([1.8830408 , 2.8070176 , 0.73099405, 1.4385962 , 0.09356717],
      dtype=float32)
```